EnergyPlus-Trained Hourly Heating and Cooling Demand Surrogates for the Vienna Building Stock

Philipp Thunshirn
Department of Industrial Engineering
University of Applied Sciences Technikum Wien
Vienna, Austria
philipp.thunshirn@technikum-wien.at

***Abstract*—This study develops an EnergyPlus-trained surrogate for hourly heating and cooling demand of the Vienna building stock. The stock comprises 134.6 million square metres of gross floor area and is represented by eight residential and nonresidential cohorts covering four construction periods. Each cohort is simulated as a reference archetype, and the resulting demand intensity is scaled by its Citiwatts floor area. Separate histogram gradient-boosting models combine demand-state classification with magnitude regression. Validation on complete calendar weeks excluded from training gives coefficients of determination of 0.959 for heating and 0.975 for cooling. After floor-area-weighted aggregation, annual city demand differs from EnergyPlus by 1.7 percent for heating and 0.8 percent for cooling. Peak demand is underestimated by 6.7 percent and 10.1 percent, respectively. The complete EnergyPlus demand path takes 67.1 seconds for the eight annual cohort simulations, whereas surrogate inference for the same 70,080 cohort-hours takes 1.73 seconds. The resulting speedup is approximately 39.**

***Index Terms*—building energy simulation, building stock, cooling demand, EnergyPlus, heating demand, surrogate model, Vienna**

I. Introduction

Hourly heating and cooling profiles are an important input to energy-system models with high shares of renewable energy. Annual demand alone does not show when heat pumps draw electricity, when heating coincides with a winter peak, or when cooling increases the summer load. These questions require a building model that responds to weather, solar and internal gains, occupancy schedules, the thermal envelope, and the operation of heating and cooling systems. Simple degree-day relations or fixed normalized profiles can reproduce an annual total, but they do not represent these interactions explicitly.

EnergyPlus provides a considerably richer description of building demand [1], [2]. It solves the transient heat balance and accounts for heat transfer through opaque surfaces and windows, ventilation and infiltration, solar radiation, internal gains, thermal storage, and HVAC controls. Heating and cooling loads therefore follow from the same physical balance instead of being imposed as independent profiles. EnergyPlus is also a widely used and extensively documented building-performance simulation program. Its accuracy still depends on appropriate geometry, material properties, schedules, and weather. Given suitable inputs, however, it provides a more defensible basis for deriving hourly building loads than a profile assembled from annual energy and outdoor temperature alone.

The detail that makes EnergyPlus useful also complicates its integration into larger models. A city-scale application requires several representative buildings, input files and schedules, repeated annual simulations, and the extraction of results from EnergyPlus output files. Direct coupling becomes particularly burdensome when building demand is evaluated repeatedly in a scenario analysis or optimization model. The building simulation can then become a computational and technical bottleneck even when one annual run is not prohibitively slow.

A surrogate model offers a practical division of work. EnergyPlus is retained offline as the detailed teacher, while a statistical model learns the mapping from weather, operating conditions, and building properties to useful heating and cooling demand. Once trained, the surrogate can be called directly by an energy-system model without constructing and executing a new EnergyPlus run [3], [4]. The gain in speed is only useful if the surrogate preserves annual energy, hourly variation, seasonal transitions, and peak demand.

This paper demonstrates the approach for the Vienna building stock. Citiwatts floor area and construction-period data are represented by eight residential and nonresidential cohorts. EnergyPlus simulations for the 2023 weather year provide hourly useful heating and cooling for each cohort, and a gradient-boosting surrogate emulates these outputs. The analysis assesses hourly accuracy on periods excluded from training, accuracy after aggregation to the Vienna stock, and computational performance. Building flexibility, dispatch, and energy-system optimization are outside the present scope.

II. Materials and Methods

A. Case Study and Building-Stock Representation

The case study covers the Vienna building stock represented in the Citiwatts inventory [5]. The inventory reports 134.6 million m^2 of gross floor area and 403.7 million m^3 of heated volume. Residential buildings account for 89.0 million m^2, and nonresidential buildings account for 45.5 million m^2. Reported annual heat consumption is 17.54 TWh. This quantity includes domestic hot water and serves as a top-down energy-balance reference; it is distinct from the useful space-heating output of the EnergyPlus simulations.

The stock was divided into residential and nonresidential buildings and four construction periods: before 1975, 1975–1990, 1990–2000, and 2000–2014. The corresponding raw shares were 54, 24, 4, and 17 percent and were normalized before application within each sector. The resulting eight cohorts sum to the complete Citiwatts floor area. They represent an aggregation of the Vienna stock rather than a sample of individual buildings.

Table I reports the floor-area weights and principal thermal parameters. Period-dependent U-values and residential geometry are informed by Austrian TABULA and EPISCOPE apartment-building data [6], [7]. The same period-dependent U-value classes were used for nonresidential buildings because an equivalent Vienna typology was unavailable. Residential window solar transmittance values were derived from comparable TABULA window classes. The nonresidential geometry, nonresidential solar transmittance, and areal heat capacities are explicit model assumptions.

Residential envelope-area ratios per unit conditioned floor area were 0.82 for walls, 0.18 for windows, 0.37 for roofs, and 0.36 for exposed floors. The respective nonresidential ratios were 0.30, 0.12, 0.30, and 0.30. Residential domestic-hot-water demand is separated in the wider Vienna data set but is not simulated or predicted here. Citiwatts defines the stock extent and supplies the floor-area weights for city aggregation. Its annual heat consumption is not a surrogate target.

B. EnergyPlus Reference Simulations

One EnergyPlus 26.1 model was prepared for each cohort [2]. The models represent equivalent stock archetypes rather than particular buildings. A conditioned reference floor area of 1000 m^2 was used in all simulations. This common area keeps the geometry at a plausible archetype scale and allows the output to be expressed as demand intensity. It does not restrict the represented stock to 1000 m^2. Envelope areas, internal gains, and air volumes were scaled consistently with the reference area.

The Citiwatts volume-to-area ratio gives a reference room height of approximately 3 m and a conditioned volume of approximately 3000 m^3. Each archetype was simulated for the 8760 hours of 2023. Outdoor temperature and solar radiation were obtained from Open-Meteo and converted to a pseudo-EPW weather file [8]. A Climate.OneBuilding file for Wien-Innere-Stadt supplied EPW header and format information, but not the hourly observations [9]. Heating and cooling setpoints were 22 and 27 °C. The simulations included envelope transmission, solar and internal gains, ventilation, infiltration, and ideal-load heating and cooling.

EnergyPlus outputs were converted to specific hourly demand $q_{c,t}^{\mathrm{EP}}$ in kWh/m^{2}h. City-level demand was obtained by multiplying the intensity by represented floor area A_c and summing across cohorts:

$$Q_{c,t} = q_{c,t}^{\mathrm{EP}} A_c, \qquad Q_t^{\mathrm{Vienna}} = \sum_{c=1}^{8} Q_{c,t}. \tag{1}$$

The procedure represents the complete inventory area through eight specific demand profiles. It does not calibrate the EnergyPlus annual sum to the 17.54-TWh Citiwatts heat balance, which contains end uses outside the EnergyPlus targets. The unnormalized EnergyPlus series is retained so that teacher and surrogate are compared on the same basis.

C. Dataset Generation and Preprocessing

Combining eight cohort simulations with 8760 time steps produced 70,080 observations. One observation describes one cohort during one hour. The two target variables are useful space-heating demand and useful cooling demand, both in kWh/m^{2}h. Separate models were trained for the two targets.

The explanatory variables comprise:

- weather: outdoor temperature and global horizontal, direct normal, and diffuse horizontal irradiation;
- operation: heating and cooling setpoints, internal gains, infiltration, and ventilation air-change rates;
- time: sine and cosine transformations of hour of day and day of year; and
- building properties: sector, cohort indicators, envelope U-values, area ratios, and areal heat capacity.

The periodic time transformations avoid artificial discontinuities at midnight and between calendar years. A specific heat-loss coefficient was calculated from the products of U-values and their corresponding area ratios. Heating and cooling temperature differences between outdoor temperature and the respective setpoint, as well as their products with the heat-loss coefficient, were added as physically meaningful predictors.

All cohort-hour combinations were retained; no random subsampling was applied. Data preparation required eight complete annual profiles, checked for duplicate cohort–timestamp pairs, and rejected missing values. Targets were normalized by represented cohort area before fitting. Floor-area values were used again only after prediction to obtain the city-level series.

D. Surrogate Model

Heating and cooling were modelled independently using the histogram-based gradient-boosting implementation in scikit-learn [10]–[12]. Gradient boosting constructs an additive ensemble of decision trees in which each iteration reduces the residual error of the current ensemble. The model class accommodates the mixture of continuous weather and envelope parameters, cyclic time variables, and categorical cohort information.

Both targets contain long periods without demand. A single regression model gave excessive influence to these hours and produced residual summer heating. A two-part hurdle formulation was therefore used for each service. A `HistGradientBoostingClassifier` first estimates the probability $\widehat{p}_t$ that demand is present. A `HistGradientBoostingRegressor` estimates demand magnitude $\widehat{q}_t^{\mathrm{reg}}$. During inference,

$$\widehat{q}_t = \begin{cases} \max(0, \widehat{q}_t^{\mathrm{reg}}), & \widehat{p}_t \geq \tau, \\ 0, & \widehat{p}_t < \tau, \end{cases} \tag{2}$$

TABLE I
VIENNA BUILDING-STOCK COHORTS AND THERMAL ARCHETYPE PARAMETERS

Construction period	Floor area (million m^2)		U-value (W/m^{2}K)				Heat capacity (Wh/m^{2}K)		g_{res}
	Residential	Nonresidential	Wall	Window	Roof	Floor	Residential	Nonresidential	
Before 1975	48.6	24.8	1.40	2.30	1.70	1.20	80	70	0.85
1975–1990	21.6	11.0	1.10	2.70	0.80	0.80	75	65	0.76
1990–2000	3.6	1.8	0.60	2.50	0.50	0.50	70	60	0.60
2000–2014	15.3	7.8	0.35	1.40	0.20	0.40	65	55	0.50
Total	89.0	45.5							

Nonresidential window solar transmittance: $g_{\mathrm{nonres}} = 0.60$ for all periods.

where τ is a service-specific threshold.

Monotonic constraints were applied where the expected direction is unambiguous. Predicted heating cannot increase with outdoor temperature and increases with the heating temperature difference. Cooling follows the opposite temperature response. The regression component used a learning rate of 0.05, up to 700 boosting iterations, at most 159 leaf nodes per iteration, and an L2 regularization coefficient of 0.01. The classifier used up to 300 iterations, 63 leaf nodes, and an L2 coefficient of 0.05.

E. Training and Validation

All 70,080 observations entered the cross-validation procedure. Sample weights prevented moderate- and zero-demand hours from dominating the regression loss. Positive observations up to the median positive demand received weight 2, whereas the highest 15 percent of positive-demand observations received weight 9. Remaining observations received weight 1. Positive and zero classes were balanced within each classifier training fold.

The hurdle threshold was selected independently in each training fold from 0.10 to 0.90 in increments of 0.05. Selection retained energy during true on-hours while penalizing predictions during physically implausible off-hours. Threshold selection used training-fold predictions only. The selected threshold was 0.10 in all heating folds and ranged from 0.65 to 0.70 in the cooling folds.

Temporal generalization was assessed by four-fold grouped cross-validation using the ISO calendar week as grouping variable. All hours of a given week and all eight cohorts were assigned to the same fold. Adjacent hours were therefore not divided randomly between training and validation data. The procedure evaluates unseen periods within 2023, but not a new building class or an independent weather year.

Held-out cohort-hour predictions were evaluated with the coefficient of determination, mean absolute error, and root mean squared error. Predictions were then scaled according to (1). Annual relative error, peak-hour relative error, and Pearson correlation were calculated for the fully out-of-fold Vienna series. Deployable models were finally refitted to the complete data set; these fits were not used to report accuracy.

F. Runtime Assessment

The EnergyPlus runtime path included schedule and IDF generation, execution of the simulation engine, and extraction of hourly results from the SQL output. Engine time was measured three times per cohort and aggregated from the cohort medians. Inference for both surrogate targets was repeated five times, and the median duration was reported. Plot generation was disabled. The one-time training duration was recorded separately and excluded from the inference speedup.

TABLE II
CALENDAR-WEEK CROSS-VALIDATION RESULTS

Metric	Heating	Cooling
Hourly R^2	0.959	0.975
MAE (kWh/m^{2}h)	5.8×10^{-4}	5.4×10^{-4}
City annual error (%)	$+1.7$	-0.8
City peak-hour error (%)	-6.7	-10.1
City Pearson r	0.982	0.989

III. RESULTS

A. EnergyPlus Cohort Response

The EnergyPlus heat balances differ markedly between residential construction periods, as shown in Fig. 1. During the selected winter day, approximate transmission losses decline from about 82 W/m^2 for the pre-1975 cohort to about 26 W/m^2 for the 2000–2014 cohort. Ventilation losses are similar because the same air-change assumption is used across cohorts. Infiltration is omitted from the figure because its common rate does not distinguish construction periods.

B. Surrogate Accuracy

Table II summarizes out-of-fold accuracy. At cohort-hour level, the surrogate reaches coefficients of determination of 0.959 for heating and 0.975 for cooling. After floor-area-weighted aggregation, annual city demand differs from EnergyPlus by 1.7 percent for heating and minus 0.8 percent for cooling. The temporal correlations are 0.982 and 0.989. Peak heating is underestimated by 6.7 percent and peak cooling by 10.1 percent.

Annual heating errors for the large residential cohorts remain within a few percent. The largest cohort-relative error occurs for nonresidential buildings from 2000–2014 at 23 percent. Their EnergyPlus heating intensity is only 3.8 kWh/m^{2}a; consequently, a small absolute deviation gives a large percentage error and has little influence on the aggregate result. Annual cooling errors remain within approximately 2 percent for all cohorts.

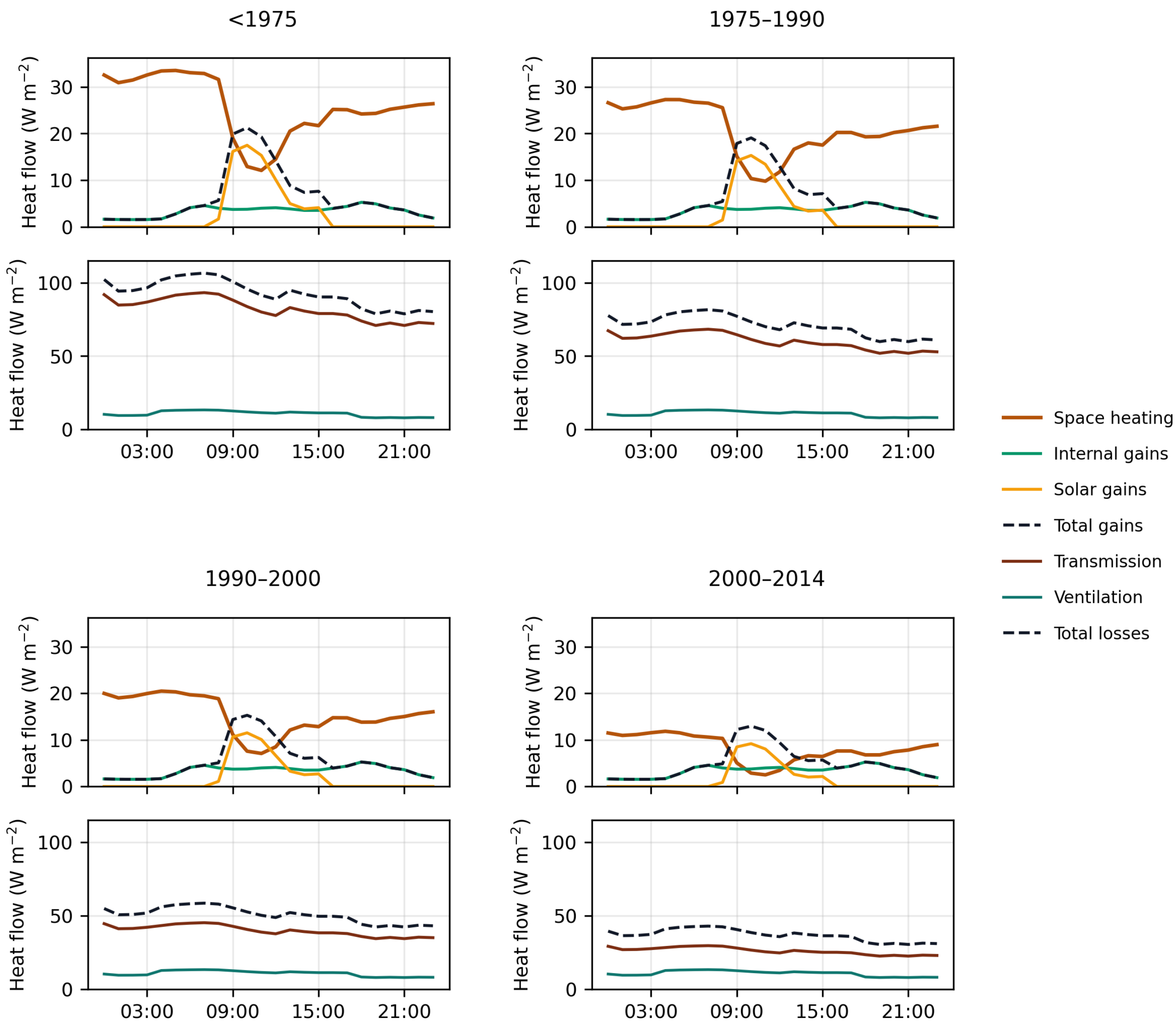

Fig. 1. EnergyPlus teacher heat-balance flows for four residential construction periods during a winter day. Upper panels show space heating, internal gains, solar gains, and total gains. Lower panels show approximate transmission and ventilation losses.

The seasonal weeks in Fig. 2 show how aggregate errors are distributed over time. During the peak-heating week, the surrogate follows the daily variation but underestimates the highest loads; total weekly energy is 1.6 percent below EnergyPlus. In the early-April heating week, energy is 3.5 percent higher and the peak is closely reproduced. Cooling profiles in late August and September agree closely in most hours, while the absolute August peak remains underestimated.

Fig. 3 relates city demand to outdoor temperature. The 1-K bin means produced by the surrogate follow those of EnergyPlus. Heating decreases as outdoor temperature rises, whereas cooling increases at high temperatures. The transition between both regimes is retained without a substantial band in which both services are simultaneously overpredicted.

EnergyPlus produces 0.43 GWh of heating from June through August, compared with 1.84 GWh from the surrogate. This residual is small relative to the approximately 5.3-TWh annual EnergyPlus space-heating series, but it shows that the demand-state transition is not reproduced perfectly.

C. Computational Performance

The complete EnergyPlus demand path takes 67.1 s, including 27.5 s for input preparation, 24.8 s for the simulation engine, and 14.8 s for SQL extraction, as shown in Table III. Surrogate inference for both targets takes 1.73 s, corresponding to a speedup of 38.8. Engine wall-clock time alone is 14.3 times the inference time. The one-time fit requires 18.5 s.

The inference comparison assumes that the hourly feature table is available; its preparation is not included. For 50

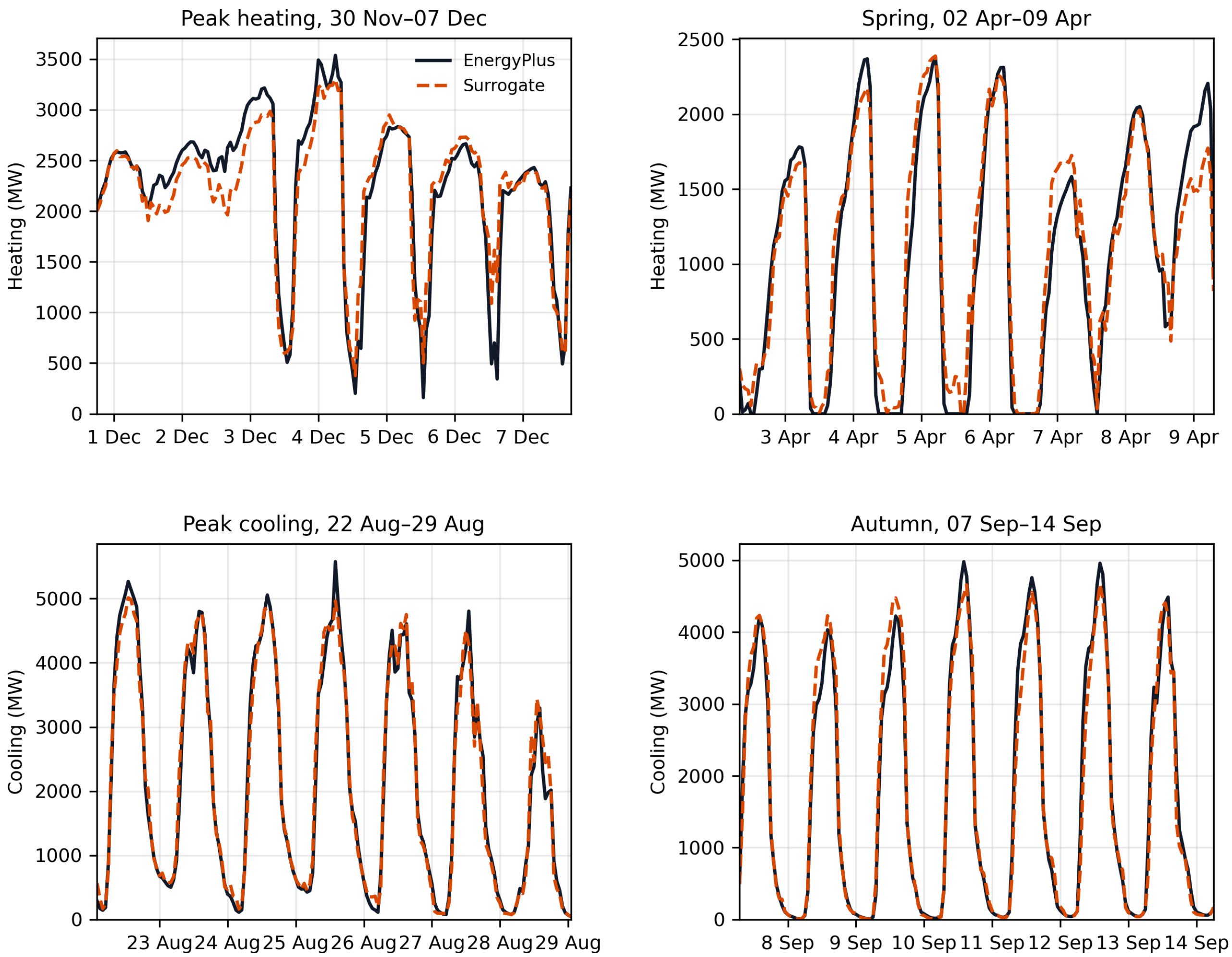

Fig. 2. Floor-area-weighted Vienna useful demand from EnergyPlus and held-out surrogate predictions. The panels show peak heating, April heating, peak cooling, and September cooling. Each interval contains 168 consecutive hours.

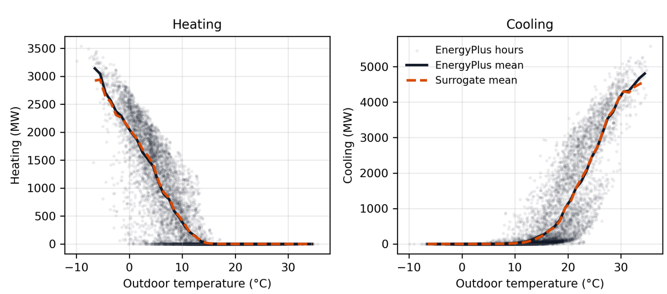

Fig. 3. Floor-area-weighted Vienna heating and cooling demand as a function of outdoor temperature. Points show hourly EnergyPlus values. Lines show EnergyPlus and surrogate means in 1-K temperature bins with at least eight observations.

repeated city-year evaluations, the difference between both paths amounts to approximately 55 min.

TABLE III
RUNTIME FOR EIGHT COHORTS AND 8760 HOURS

Path	Prep. (s)	Engine (s)	SQL (s)	Total (s)
EnergyPlus	27.5	24.8	14.8	67.1
Surrogate inference	–	–	–	1.73

IV. Discussion

The results show that a compact statistical model can retain most of the hourly variation generated by the EnergyPlus cohort simulations. Annual energy and temporal correlation are reproduced more accurately than maximum load. The surrogate is therefore suitable for producing hourly demand shapes that are subsequently normalized to an independently specified annual energy balance. It should not be used without qualification when the EnergyPlus maximum itself is treated as a design load.

The city result covers all floor area in the Citiwatts inventory, but it does not resolve every building in Vienna. Buildings within one sector and construction period share an equivalent archetype and operating profile. Aggregation retains major differences in envelope quality while suppressing variation in geometry, occupancy, refurbishment, and control behaviour within each cohort. Nonresidential archetypes carry additional uncertainty because Vienna-specific geometry and window data were unavailable at the same level as for Austrian residential apartment buildings.

Validation excludes complete calendar weeks from training, but remains limited to one weather year and the same eight cohorts. The reported accuracy supports interpolation across unseen periods of 2023, not transfer to a different climate, an extreme year outside the training range, or an unseen building type. The remaining peak-load underestimation is consistent with this limitation.

The surrogate reproduces reference heating and cooling demand only. It does not contain setpoint changes, preheating, load curtailment, rebound, or indoor-temperature state transitions. These processes require a separate stateful response model before building flexibility can be represented in an optimization problem.

V. Conclusion

An EnergyPlus-to-surrogate workflow was developed for hourly useful heating and cooling demand of the Vienna building stock. Eight archetypes represent the complete Citiwatts floor area across two sectors and four construction periods. Their EnergyPlus outputs form a 70,080-row annual data set. Separate histogram gradient-boosting hurdle models reproduce held-out cohort-hour demand with coefficients of determination of 0.959 for heating and 0.975 for cooling.

After aggregation to the Vienna stock, annual errors remain below 2 percent, whereas peak loads are underestimated by 7 to 10 percent. Surrogate inference is approximately 39 times faster than the complete EnergyPlus demand path. The model therefore provides an efficient source of EnergyPlus-consistent demand profiles for subsequent energy-system analyses, provided that annual energy anchors and peak-load limitations remain explicit.

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